

CodeAlpha Internship – Task 1

Web Scraping Project Submission

Project: Book Data Web Scraper

Website: Books to Scrape (public practice website)

1. Objective

The objective of this task is to build a Python web scraper that collects useful information from a public website and saves the collected information as a custom dataset.

2. Tools and Technologies

Tool / Technology	Purpose
Python	Programming language used for scraping
Requests	Sends HTTP requests to the website
BeautifulSoup4	Parses HTML and extracts data
CSV	Stores the collected data as a dataset
VS Code	Development environment

3. Data Collected

The scraper extracts the following fields from the books website:

Field	Description
Title	Name of the book
Price	Listed book price
Rating	Star rating
Availability	Whether the book is in stock
Product URL	Link to the book page

4. How the Scraper Works

1. Sends a request to the website.
2. Reads the returned HTML using BeautifulSoup.
3. Finds book cards using HTML/CSS selectors.
4. Extracts title, price, rating, availability and URL.
5. Follows the Next-page link until all pages are processed.

6. Saves the collected records into **books_dataset.csv**.

5. Main Python Code

```
import requests
from bs4 import BeautifulSoup

response = requests.get(url)
soup = BeautifulSoup(response.text, "html.parser")

for book in soup.select("article.product_pod"):
    title = book.select_one("h3 a")["title"]
    price = book.select_one(".price_color").get_text(strip=True)
    rating = book.select_one("p.star-rating").get("class")[-1]
    # save extracted data to CSV
```

6. Output Dataset

The project creates a CSV file named **books_dataset.csv**. The CSV contains the scraped book records and can be opened in Excel or Google Sheets.

7. Project Files

scrape_books.py – web scraper code
requirements.txt – required Python libraries
README.md – project instructions and explanation
books_dataset.csv – collected dataset

8. Conclusion

A Python-based web scraper was developed using Requests and BeautifulSoup. It extracts book information from a public practice website, handles pagination, and stores the collected information in a structured CSV dataset. This demonstrates basic web scraping, HTML parsing, data extraction and dataset creation.